

SOFTWARE ENGINEERING

LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Project: Food Bank Surplus Redistribution Platform

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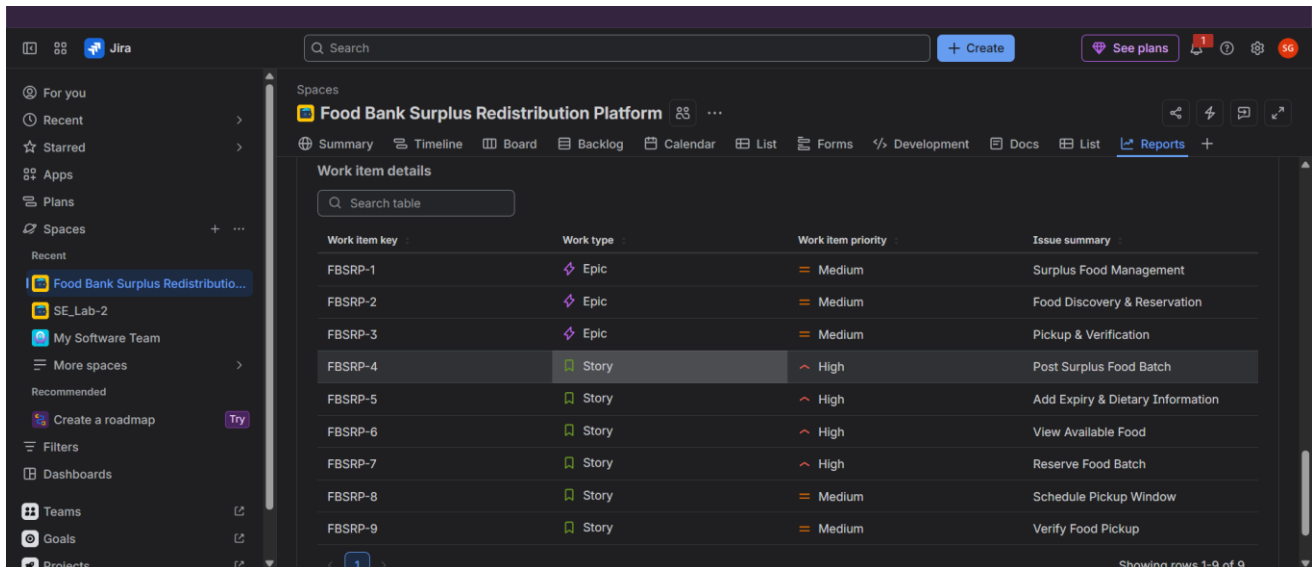
Objective

To use Jira to convert the Food Bank Surplus Redistribution Platform requirements into Agile Epics and User Stories, prioritize and estimate the backlog using Story Points, simulate two Scrum sprints, and analyse sprint progress.

1. EPICS

Epic	Description
Epic 1: Surplus Food Management	Manage surplus food batches, including posting food and recording expiry and dietary information.
Epic 2: Food Discovery & Reservation	Allow shelters to discover available food and reserve suitable surplus food batches.
Epic 3: Pickup & Verification	Support pickup scheduling and verification of successful food collection.

Jira evidence: the work-item details view shows the three Epics and six Stories created for the project.



Work item key	Work type	Work item priority	Issue summary
FBSRP-1	Epic	Medium	Surplus Food Management
FBSRP-2	Epic	Medium	Food Discovery & Reservation
FBSRP-3	Epic	Medium	Pickup & Verification
FBSRP-4	Story	High	Post Surplus Food Batch
FBSRP-5	Story	High	Add Expiry & Dietary Information
FBSRP-6	Story	High	View Available Food
FBSRP-7	Story	High	Reserve Food Batch
FBSRP-8	Story	Medium	Schedule Pickup Window
FBSRP-9	Story	Medium	Verify Food Pickup

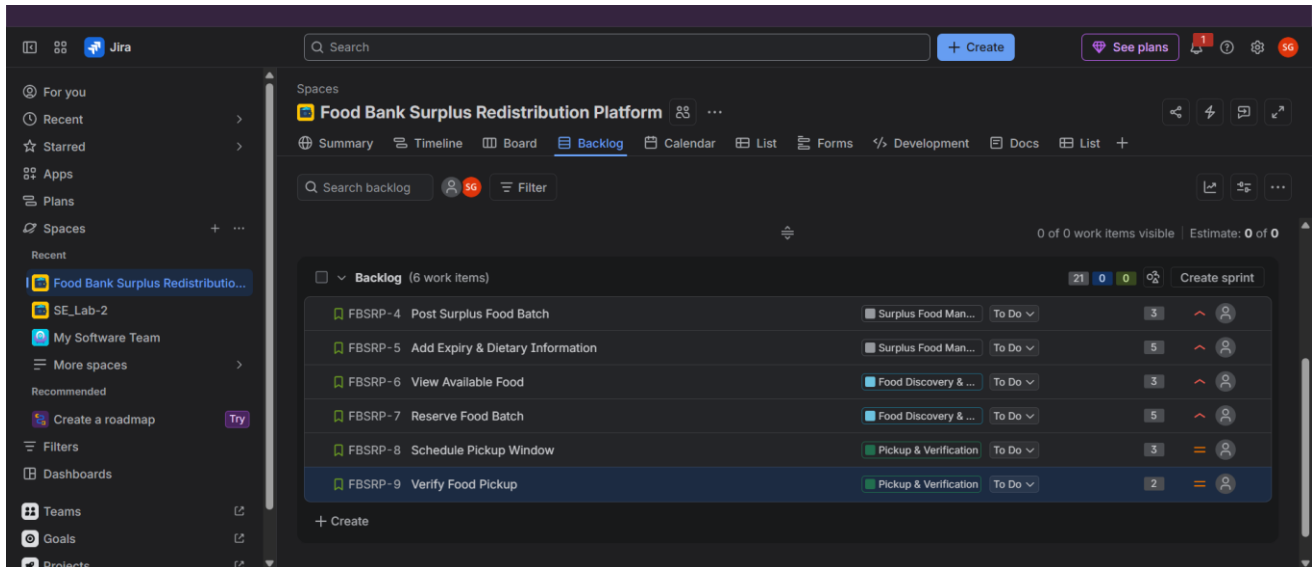
2. User Stories

Key	User Story	Epic	User-story statement
FB SRP-4	Post Surplus Food Batch	Surplus Food Management	As a donor partner, I want to post a surplus food batch so that shelters can discover available food.
FB SRP-5	Add Expiry & Dietary Information	Surplus Food Management	As a donor partner, I want to add expiry and dietary information so that shelters can select suitable food safely.
FB SRP-6	View Available Food	Food Discovery & Reservation	As a shelter coordinator, I want to view available food batches so that I can find suitable surplus food.
FB SRP-7	Reserve Food Batch	Food Discovery & Reservation	As a shelter coordinator, I want to reserve a food batch so that the surplus food is allocated for pickup.
FB SRP-8	Schedule Pickup Window	Pickup & Verification	As a shelter coordinator, I want to schedule a pickup window so that collection can be organized efficiently.
FB SRP-9	Verify Food Pickup	Pickup & Verification	As a shelter coordinator, I want to verify food pickup so that successful collection is recorded.

3. Prioritization and Story Point Estimation

Work Item	Summary	Priority	Story Points
FB SRP-4	Post Surplus Food Batch	High	3
FB SRP-5	Add Expiry & Dietary Information	High	5
FB SRP-6	View Available Food	High	3
FB SRP-7	Reserve Food Batch	High	5
FB SRP-8	Schedule Pickup Window	Medium	3
FB SRP-9	Verify Food Pickup	Medium	2

Total estimated backlog: 21 Story Points. The estimates use small Fibonacci-style values and reflect relative effort, complexity, and uncertainty.



4. Sprint 1

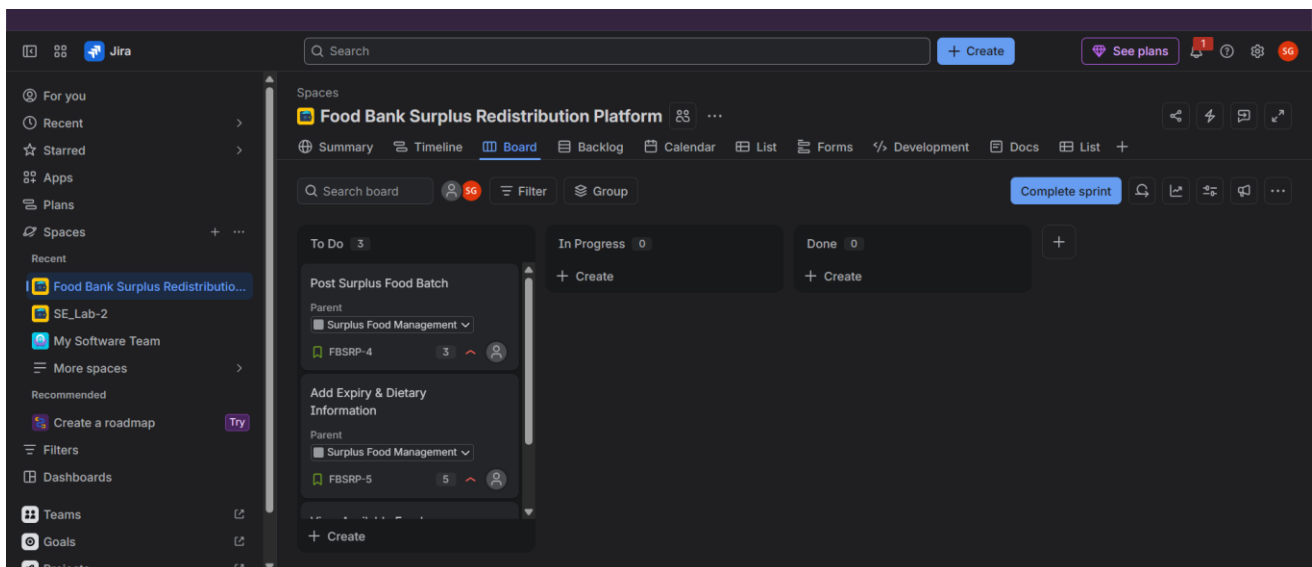
Sprint Duration: 1 Week

Sprint Goal: Enable shelters to discover available surplus food and begin the food redistribution process.

Stories completed:

- Post Surplus Food Batch — 3 points
- Add Expiry & Dietary Information — 5 points
- View Available Food — 3 points

Sprint 1 total: 11 Story Points.



5. Sprint 2

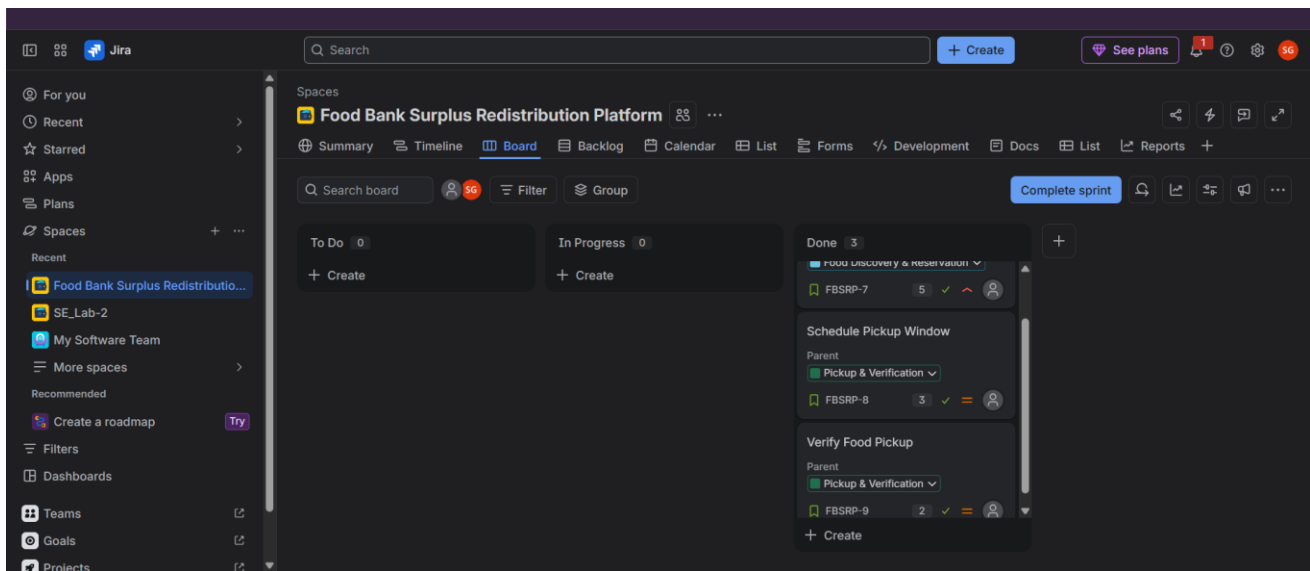
Sprint Duration: 1 Week

Sprint Goal: Enable shelters to reserve food, schedule pickup, and verify successful collection.

Stories completed:

- Reserve Food Batch — 5 points
- Schedule Pickup Window — 3 points
- Verify Food Pickup — 2 points

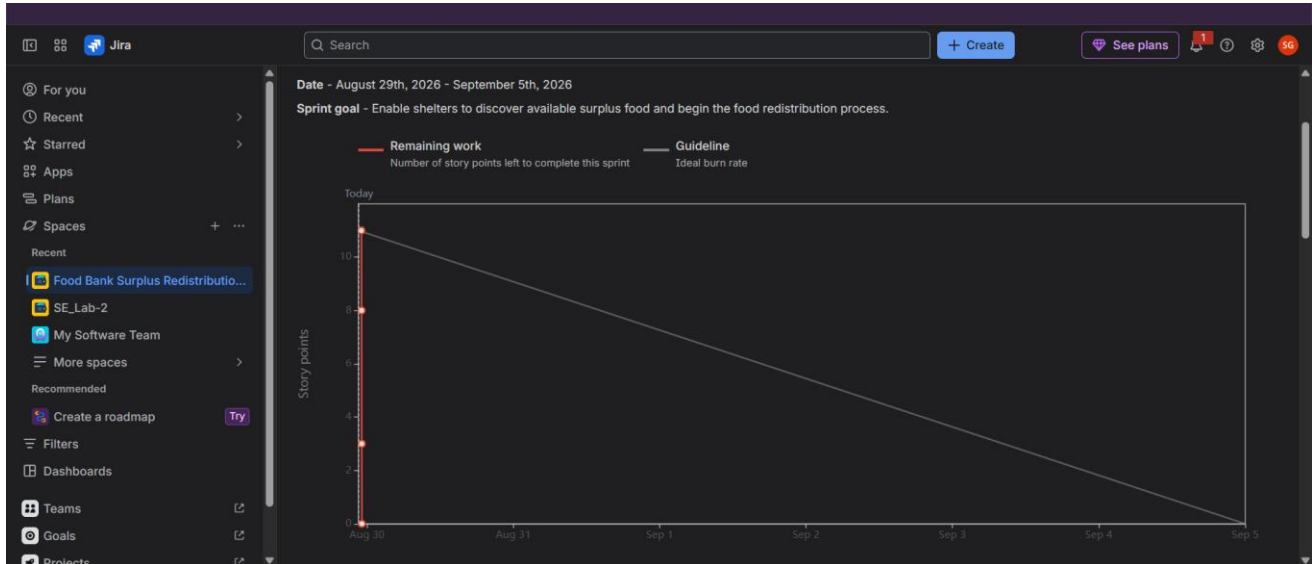
Sprint 2 total: 10 Story Points.



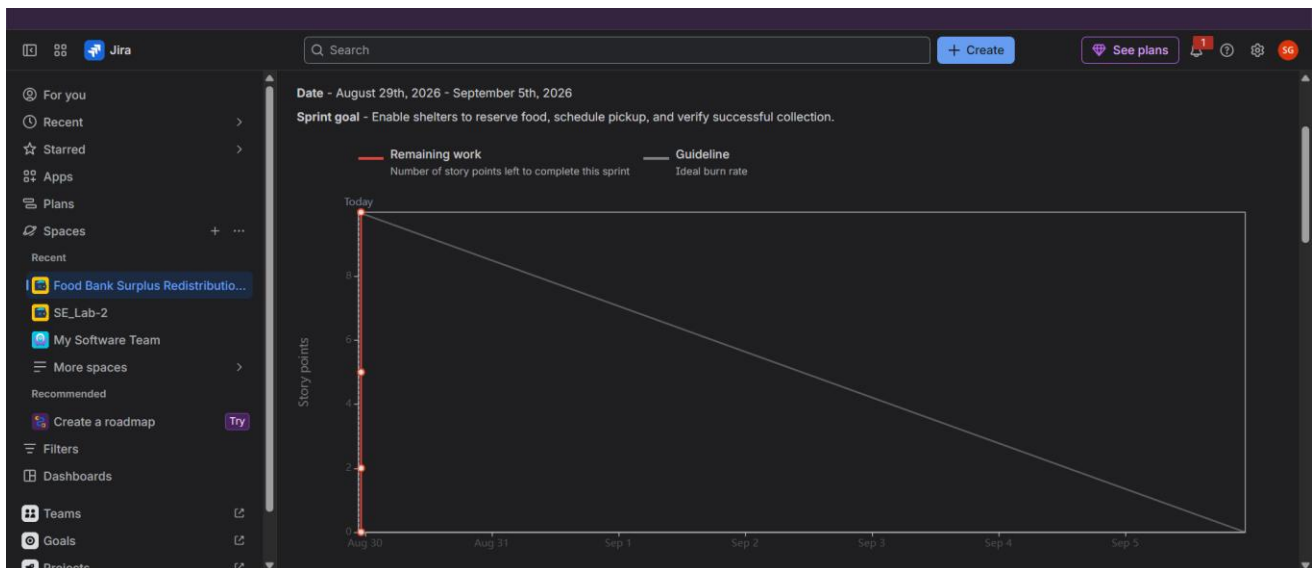
6. Burndown Charts

The Jira Burndown Chart represents the amount of work remaining in the sprint. The red line shows remaining Story Points, while the grey guideline represents the ideal linear burn rate.

Sprint 1 Burndown Chart



Sprint 2 Burndown Chart



7. Reflection

1. Did your estimations reflect the actual effort?

The Story Point estimates provided a reasonable relative measure of effort for the selected features. The larger 5-point items represented more involved work than the 2- and 3-point items. Because the activity was a simulated sprint, the estimates mainly helped compare relative complexity.

2. Was your backlog well-prioritized?

Yes. Core food-posting and food-discovery functions were given High priority because they form the main flow of surplus-food redistribution. Pickup-related supporting functions were assigned Medium priority.

3. How did your simulated sprint align with your plan?

The simulated work followed the planned backlog order. Sprint 1 completed the first three stories for 11 Story Points, and Sprint 2 completed the remaining three stories for 10 Story Points. Stories were moved through the Scrum workflow from To Do to In Progress to Done.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Charts showed that the planned work was completed within each simulated sprint. The sharp drop in remaining work reflects that the simulation was completed quickly rather than gradually across the full week. The charts therefore confirm completion and also illustrate why a more realistic simulation would spread work across multiple days.

8. Conclusion

The Food Bank Surplus Redistribution Platform requirements were successfully organized in Jira using Agile Scrum practices. Three Epics and six User Stories were created, prioritized, and estimated for a total of 21 Story Points. Two one-week simulated sprints were planned and completed, with Sprint 1 delivering 11 points and Sprint 2 delivering 10 points. The Jira Burndown Charts reached zero remaining work for both sprints, demonstrating the use of Jira for backlog management, sprint planning, task tracking, and sprint progress analysis.